Build a Calculator with Claude

30-Minute Group Activity | Python | Replit + claude.ai

Your Goal

By the end of this activity your group will have a working Python calculator that can add, subtract, multiply, and divide. You don't need to know how to code yet. Claude will be your coding partner: you'll ask it questions, get code, run it, test it, and fix problems together.

Before You Start

What you need

- **Groups of 3.** One laptop runs Replit, one laptop runs claude.ai (or use split screen on one laptop).
- **A Replit account** (replit.com) and **a claude.ai account**.
- This handout, a pen, and a willingness to break things and fix them.

Group roles

Pick one role each. Switch roles at the stretch challenge (minute 22) so everyone tries something new.

Role	What you do
Prompter	Types messages to Claude. Reads Claude's answers out loud to the group.
Coder	Controls Replit. Pastes code, clicks Run, types answers into the program.
Tester / Recorder	Chooses test numbers, fills in the Test Log, and watches for mistakes.

Activity Timeline

Time	Step	What happens
0:00 – 0:03	1. Set up	Form groups, pick roles, open Replit and Claude.
0:03 – 0:08	2. Plan with Claude	Ask Claude what a calculator program needs.
0:08 – 0:15	3. Build	Get the code from Claude and run it in Replit.
0:15 – 0:22	4. Test & fix	Try every operation, record results, fix errors.
0:22 – 0:27	5. Stretch challenge	Switch roles and add one new feature.
0:27 – 0:30	6. Share & reflect	Demo to another group and answer reflection questions.

Step 1: Set Up (3 min)

1. Go to **replit.com** and log in.
2. Click **Create App** (it may say **+ Create** or **Create Repl**). If Replit asks what you want to build with AI, choose the option to start from a **template** instead.
3. Choose the **Python** template and name it **group-calculator**.
4. You should see a file called **main.py** and a green **Run** button. That's where your code goes.
5. In another tab or laptop, open **claude.ai** and start a new chat.

> **Heads up**
> Use Claude for this activity, not Replit's built-in AI Agent. If Replit offers to write code for you, close that panel.

Step 2: Plan with Claude (5 min)

Good programmers plan before they code. The Prompter types this into Claude:

> **Prompt 1**
> We're beginners and have never coded before. We want to build a simple calculator in Python that asks the user for two numbers and an operation (+, -, *, /), then prints the answer. Before writing any code, explain in plain English the steps the program needs to follow. Keep it short.

Read Claude's answer out loud. As a group, write the steps in your own words:

1. _____
2. _____
3. _____
4. _____

Vocabulary check: If Claude uses a word you don't know (like *variable*, *input*, or *if statement*), ask it: *"What does ___ mean? Explain it like I'm new to coding."*

Step 3: Build (7 min)

Now ask Claude for the code. The Prompter types:

> **Prompt 2**
> Great. Now write that calculator in Python. Keep it simple enough for beginners, put a comment above each part explaining what it does, and make it work in Replit.

1. Claude will show a code block. Click the **Copy** button on the code block.
2. The Coder deletes anything already in **main.py** and pastes the code.
3. Click **Run**. The program will ask questions in the **Console** (or **Shell**) panel. Click in that panel and type your answers, pressing Enter after each.
4. Before moving on, the whole group reads the comments in the code. Point at the line that does the adding. Can you find it?

Step 4: Test & Fix (7 min)

Run the program once for each row below. Click Run again each time. The Tester fills in what actually happened.

#	First number	Operation	Second number	Expected	What we got	Pass?
1	8	+	5	13		
2	20	-	7	13		
3	6	*	4	24		
4	15	/	3	5		
5	7	/	0	?		
6	hello	+	2	?		
7	(your pick)					

Tests 5 and 6 might crash the program. That's on purpose! When it crashes, Replit shows a red error message. Fix it with Claude:

> **Prompt 3 (fixing bugs)**
> Our calculator crashed when we [describe what you did]. Here is the error message: [paste the red error text]. Explain what went wrong in simple terms and give us the fixed code.

Paste the new code into main.py, click Run, and try the test again. Mark it as a pass once it works.

Step 5: Stretch Challenge (5 min)

Switch roles now. Then pick ONE feature and ask Claude to add it to your existing code:

- **Keep going:** the calculator asks "Do another calculation? (y/n)" instead of ending.
- **More math:** add powers (**) or remainders (%).
- **Friendly errors:** if someone types an unknown operator, show a helpful message.
- **Your idea:** anything your group thinks would make it better.

> **Prompt 4**
> Here is our current calculator code: [paste your code]. Add this feature: [describe it]. Show us the full updated code and point out which lines are new.

Step 6: Share & Reflect (3 min)

Pair up with another group. Each group runs its calculator and shows off the stretch feature. Then answer together:

1. What was one thing Claude explained that helped you understand the code?
2. Did Claude ever give you code that didn't work right the first time? How did you find out?
3. Why is it important to test code yourself instead of just trusting the AI?
4. What's one prompt you'd write differently next time?

Tips for Working with Claude

- **Say you're a beginner.** Claude will explain things more simply.
- **Be specific.** "It doesn't work" is hard to fix. "It crashed when I typed 0 as the second number" is easy.
- **Always paste the error.** The red error text has the clues Claude needs.
- **Ask why.** If you don't understand a line, ask Claude to explain just that line.
- **You're the boss.** Claude can make mistakes. Your tests are how you catch them.

Instructor Notes

Prep checklist

- Confirm every student (or at least one per group) can log in to Replit and claude.ai before class.
- Do a quick 1-minute demo of where the Run button and Console panel are in Replit.
- Keep a timer visible and call out each step change.

What to watch for

- **Pasting in the wrong place:** students may paste Claude's explanation text along with the code. Only the code block goes in main.py.
- **Typing in the wrong panel:** answers to the program go in the Console, not in main.py.
- **Tests 5 and 6:** dividing by zero raises ZeroDivisionError and typing text raises ValueError. These are the key teaching moments, so let groups discover them before helping.
- **Groups finishing early:** have them try a second stretch feature or ask Claude to explain every line of their code.

Sample working solution

A basic version groups might end up with (before stretch features):

```
# Simple Calculator
print("Simple Calculator")

# Ask the user for two numbers and an operator
num1 = float(input("Enter the first number: "))
operator = input("Enter an operator (+, -, *, /): ")
num2 = float(input("Enter the second number: "))

# Do the math based on the operator
if operator == "+":
    result = num1 + num2
elif operator == "-":
    result = num1 - num2
elif operator == "*":
    result = num1 * num2
elif operator == "/":
    if num2 == 0:
        result = "Error: you can't divide by zero"
    else:
        result = num1 / num2
else:
    result = "Error: unknown operator"

# Show the answer
print("Result:", result)
```

Note: this version still crashes on test 6 (typing "hello"). Handling that requires *try/except*, which groups can add with Claude's help as a stretch goal.

Quick assessment

A group has succeeded if it can: run the calculator for all four operations, show a completed Test Log with at least one bug found and fixed, and explain in their own words what one section of the code does.